

Lower Mustang- Jomsom|~2,720–2,760m

General Overview

Planetary Health Transect

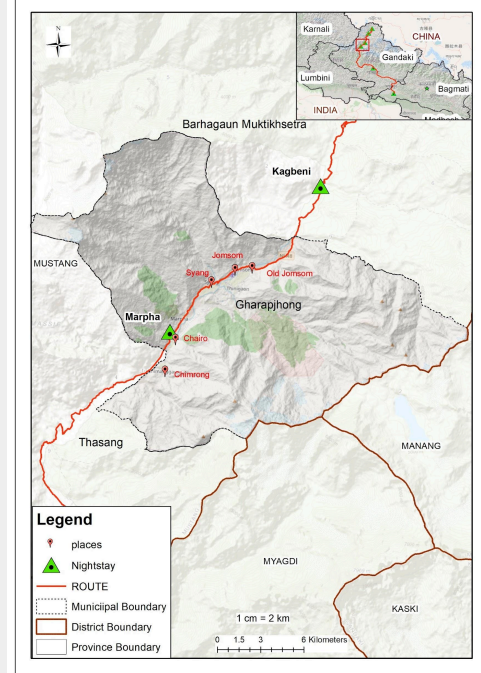
Landscape

Geography & Landscape

Jomsom (Tibetan: Dzong Sampa — 'New Fort') sits at 2,720–2,760m in the Kali Gandaki River valley, the district headquarters of Mustang and the principal gateway to Upper Mustang and Muktinath. It is bisected by the Kali Gandaki River — which forms the deepest river gorge on Earth — with the older western settlement (Old Jomsom) and the newer eastern settlement (New Jomsom) on opposing banks, connected by bridges.

The town is framed by two of the world's great peaks: Dhaulagiri (8,167m) to the west and Nilgiri (7,061m) to the east. The Kali Gandaki gorge channels a powerful thermal wind from south to north every afternoon — the defining ecological force of Jomsom, shaping architecture, agriculture, aviation, and daily life simultaneously. Traditional Thakali stone houses are engineered with low profiles and internal courtyards to deflect these winds. The sacred Dhumba Lake, fed by Nilgiri glacial meltwater, sits 5km north of town.

The landscape is an arid rain-shadow desert at altitude — dry, rocky, and windswept, with a cold semi-arid (BSk) Köppen classification and annual precipitation ~260–323mm concentrated in the July–August monsoon window. The gorge is the world's premier Shaligram ammonite fossil site, with 200-million-year-old Tethys Sea fossils embedded in the black riverbed shales — sacred to Hindu and Buddhist pilgrims since antiquity.



Demographics & Communities

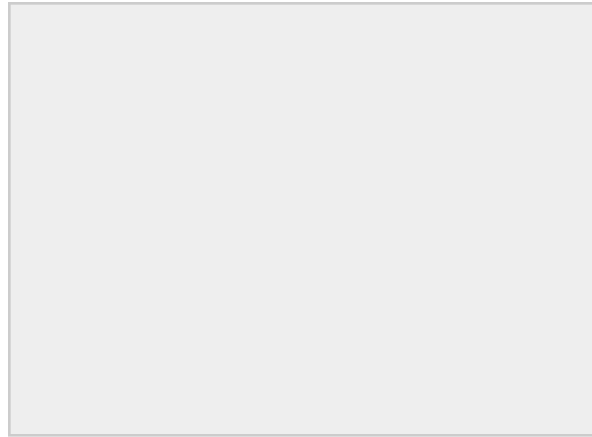
Key Statistics

- Population: 1,370 in Jomsom town (2011 census); Gharapjhong Rural Municipality total ~13,534 (2021 census); Mustang District 14,452 (2021)
- Jomsom is Ward No. 4 of Gharapjhong Rural Municipality, established 2017 — merging former Jomsom, Marpha, and Tukucho VDCs across 319.77 km²
- Ethnicity: predominantly Thakali (Mawatan sub-group — Panchgaon Thakali); with Gurung, Magar, Tibetan-Bhote in-migrants and government official postings
- Religion: predominantly Buddhist (~75% district); Thakali also observe Hindu festivals; Bon practice adjacent (Jomsom Bon Monastery, affiliated to Lubra Gompa)



Aerial view of Jomsom straddling the Kali Gandaki River

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- Out-migration: >80% of Thakali people now reside outside Thak Khola — in Kathmandu, Pokhara, India, and abroad; diaspora drives remittance economy
 - Annual growth rate: declining due to net out-migration; seasonal population swings dramatically with trekking season (Oct–Nov, Mar–May) and pilgrimage surges
 - Literacy rate: ~70% (district level); female literacy lower due to seasonal mobility patterns
 - Primary occupation: government services, tourism/hospitality, trade, agropastoralism; Jomsom holds all district-level offices



Peoples & Ethnicity

Thakali — Panchgaon sub-group: Mawatan Thakali from the five-village Panchgaon cluster (Marpha, Chhairo, Chimang, Syang, Thini — of which Jomsom is now a sixth). Four clans: Hirachan, Lalchan, Jwarchan, Pannachan. Historically dominant trans-Himalayan salt–grain traders; now the town's core hospitality, administrative, and commercial families.

Thini sub-group: 30 minutes walk from Jomsom, Thini is the oldest Panchgaon village — ruled by King Thang Mig Chen, who built the original fort (Dzong Sampa). The Tingaonle Thakali (Thini, Syang, Chimang) consider themselves distinct from Marpha and Chhairo Thakalis and do not inter-marry with them. Thini Bhadra festival and Bhomo Ama festival maintain distinct cultural identity.

In-migrant & government cohort: Gurung, Magar, Brahmin–Chhetri, lowland Nepali — arrived post-road access (2008–2015) and as government postings (police, army, health, education, district administration). Nepal Army company stationed here. Rotating government officials from lower districts form a significant share of Jomsom's working population.

Note: Jomsom's dual identity — intimate Thakali village community and a busy government/tourism town — creates social complexity. The original Thakali families who remain maintain extraordinary hospitality traditions: Thakali cuisine (the celebrated 'Thali') draws food pilgrims alongside mountain trekkers. More than 80% of Thakalis have migrated out; those who remain often manage guesthouses, orchards, or hold district government roles.

Historical Context — Key Events

Period	Event
~12th c.	King Thang Mig Chen of Thini builds Dzong Sampa ('New Fort') to monitor north–south trade movement along the Kali Gandaki. The fort and surrounding settlement gradually become known as Jomsom — the administrative and military checkpoint between Tibetan highland and Nepalese hill cultures.
Late 18th c.	Mustang incorporated into Nepal under the Shah dynasty (1786). The Thakali emerge as dominant trans-Himalayan trade brokers, leveraging control of the Kali Gandaki salt route. Tukuche becomes the regional trade capital; Jomsom serves as administrative hub. Peak period of Thakali commercial influence — salt-for-grain exchanges with Tibet build significant prosperity in all Panchgaon villages.
1950s–60s	Tibet border closed following Chinese occupation — trans-Himalayan salt trade collapses. Jomsom and all Panchgaon settlements lose their primary income source. Mass Thakali out-migration to Pokhara and Kathmandu begins. The community shifts to agropastoralism, early tourism, and remittances. Mustang remains politically restricted; Jomsom is the last open-access settlement before the restricted zone.
Early 1970s	Jomsom becomes the official district headquarters of Mustang. All government offices — District Administration, police, army, education, health — establish here. Jomsom Airport (IATA: JMO) built, opening the first air link to Pokhara. The airport fundamentally changes Jomsom's role: from trade checkpoint to administrative and logistics hub for the entire district.
1992	Mustang opened to controlled international trekking. Jomsom Airport becomes the primary entry and exit point for trekkers on the Annapurna Circuit and Upper Mustang treks. ACAP permit office established. Annapurna Conservation Area Project (ACAP) headquartered nearby. Mustang Eco Museum founded, showcasing Buddhist monastery, herbarium, animal specimens, and traditional dress.
27 Feb 19708 Nov 199321 Aug 1998	Early aviation disasters on the Pokhara–Jomsom route establish it as one of Nepal's most hazardous air corridors: 1970 — DHC-6 crash on takeoff at Jomsom (1 killed); 1993 — Nepal Airways Y-12 runway overrun into Kali Gandaki River (no fatalities); 1998 — Lumbini Airways DHC-6 impacts Mt. Annapurna near Ghorepani on Jomsom–Pokhara flight, all 18 on board killed. Pattern of CFIT (Controlled Flight Into Terrain) in deteriorating weather established early.
14 May 2012	Agni Air Dornier 228 (Flight CHT) crashes into hillside near Marpha, 5km southwest of Jomsom Airport, killing 15 of 21 on board including Indian child actress Taruni Sachdev. Investigation identifies windshear and subsidence flow over the Kali Gandaki valley as key contributing factors alongside pilot error. The crash focuses international attention on Jomsom's aviation hazard environment.
24 Feb 2016	Tara Air Flight 193 (DHC-6-400, 9N-AHH) disappears en route Pokhara–Jomsom. Wreckage found at 10,982ft on hillside near valley entrance — all 23 on board killed. Investigation finds crew flew in IMC on VFR plan; EGPWS warnings sounded repeatedly on prior flight; no documented escape route existed for Jomsom-bound flights unable to land. Jomsom airport's lack of instrument approach procedures confirmed as systemic risk.
2008–2015	Jomsom–Kagbeni road (2008); full Beni–Jomsom–Kora La highway (2015). Journey from Pokhara reduced from 4-day trek to under 1-day drive. Domestic pilgrimage tourism to Muktinath surges to 200,000+/yr — Jomsom is the transit hub. Rapid expansion of hotels,

	restaurants, concrete construction. Gharapjhong Rural Municipality formed (2017) incorporating Jomsom, Marpha, and surrounding villages.
29 May 2022	Tara Air Flight 197 (DHC-6, 9N-AET): 22 people on board (19 passengers, 3 crew) lost contact 12 minutes after departure from Pokhara, en route Jomsom. All 22 killed; wreckage found on Sanusare Mountain, Thasang, 7.7nm SW of Jomsom Airport. TAWS inhibited by crew for entire flight; CFIT into terrain. 62-year-old captain, junior co-pilot on only their second sector flight. Second Tara Air CFIT fatality on this exact route in 6 years.
2015Earthquakes	Gorkha earthquake (April 2015) and major aftershock (May 2015) caused 15–20 Jomsom riverside buildings to suffer progressive slope failure and damage. Airport operations temporarily suspended. Restoration and slope stabilisation ongoing. District hospital and government buildings required structural assessment.

Climate Trends (1970–2019)

Jomsom sits at the boundary of the Trans-Himalayan Arid Zone — one of Nepal's fastest-warming climate regions, with precipitation showing a locally increasing trend unlike the broader Mustang decline

- Temperature rising at $\sim +0.07^{\circ}\text{C}/\text{year}$ at Jomsom station (1990–2020, Gandaki Province study); Nepal national average $0.056^{\circ}\text{C}/\text{yr}$ — nearly 3× global average ($0.02^{\circ}\text{C}/\text{yr}$)
- Annual precipitation at Jomsom: $\sim 260\text{--}323\text{mm}$ average; station data show a significant INCREASING trend of $+2.42\text{mm}/\text{year}$ (1990–2020) — unlike the declining trend at Upper Mustang ($\sim -3\text{mm}/\text{yr}$). Attributed to collision of warming monsoon westerlies with Trans-Himalayan systems
- Temperature range: January min -14°C to -12°C ; summer max $16\text{--}17^{\circ}\text{C}$; diurnal range $>20^{\circ}\text{C}$ daily — extreme thermal stress on built heritage, bodies, and equipment
- Wind: the signature climate feature — southerly Kali Gandaki valley winds accelerate daily from $\sim 10\text{am}$, routinely exceeding aviation operating limits; airport closes by 11am–noon; residents and architecture engineered around this diurnal cycle
- Winter warming faster than summer — reduced snowpack on Dhaulagiri and Nilgiri reduces Kali Gandaki baseflow, threatening Dhumba Lake recharge and Shaligram fossil bed accessibility
- High UV: 2–3× sea-level intensity; important health hazard for outdoor communities and the hundreds of thousands of pilgrims bathing at open-air Muktinath
- Key new risk: increasing precipitation variability — more intense monsoon events increasing landslide risk in geologically fragile Spiti shale terrain upstream

Residents and trekking guides directly observe warming winters, stronger pre-monsoon dust storms, and more intense summer rain events. The Kali Gandaki's afternoon wind pattern — always strong — is perceived as increasingly extreme in peak season. Jomsom's position as the district's logistical lifeline means that climate-driven airport closures cascade into medical emergency delays, supply chain disruptions, and economic loss across all of Mustang. (Gandaki Province Climate Study, IWA Publishing 2023; ICIMOD 2023)

Jomsom

Planetary Health Factsheet

Planetary Health Transect

Biodiversity

Fauna (Animal)

- ~150 bird species in the Kali Gandaki corridor; ~28 mammal species in Mustang District
- Snow Leopard, Himalayan Wolf, Blue Sheep (Bharal), musk deer, rabbit — present in forested slopes above Jomsom toward Syang and Thini; Dhumba Lake hosts migratory waterfowl
- Lammergeier, Himalayan Griffon Vulture — Kali Gandaki gorge is a globally significant raptor migration highway
- House Crow advancing upward as biological climate signal (+136m/yr recorded in Mustang 1999–2011)
- Snow Trout (Schizothorax) in Kali Gandaki declining due to GLOF-driven sediment pulses and altered hydrology — an important food source and ecological health indicator
- Kali Gandaki Shaligram ammonite fossil beds: 200-million-year Tethys Sea geology; world's sole source of these sacred stones; visible in riverbed between Jomsom and Kagbeni
- Forest patches near Jomsom (Syang, Thini area): *Pinus wallichiana*, *Betula utilis*, *Hippophae rhamnoides*, *Rhododendron* — important wildlife habitat; forest cover ends near Kagbeni



Flock of Sheep

Crops

Barley, buckwheat, potato, apple, turnip. Single cycle: Apr–Sep.

Farm Structure

Smallholdings <0.5ha. Subsistence + tourism-linked commercial. Apple products key.

Climate & Access

Wind disruption daily. Hailstorms & drought cited by >60% of farmers.

Flora (Plant)

- Mixed broad-leaved forest south of Jomsom toward Marpha: *Pinus wallichiana*, *Betula utilis*, *Hippophae salicifolia*, *Rhododendron* — highest forest cover in Mustang District
- Sub-desert scrubland at and north of Jomsom: *Caragana*, *Artemisia*, *Ephedra* — same cold-arid assemblage as Upper Mustang; forest cover ends near Kagbeni
- Irrigated oasis vegetation on Kali Gandaki terraces: barley, buckwheat, potatoes, apple orchards; seasonal *Hippophae* (sea buckthorn) along riverbanks

Agriculture

- Single crop cycle: barley–buckwheat–potato (Apr–Sep); apple and apricot as supplementary cash crops
- Smallholdings <0.5ha; mix of subsistence farming and commercial production for Jomsom's tourism market
- Traditional kar-nak crop rotation (naked barley / fallow) maintains soil fertility in nutrient-poor rocky soils
- Animal husbandry: yaks, goats, sheep, cows — dairy (butter, churpi), wool, and occasional meat; livestock housed on ground floors in winter

- Dhumba Lake aquatic flora; juniper woodland above 4,000m; alpine meadows above 4,500m toward Muktinath
- Invasive species pressure increasing from south as road and vehicle traffic brings seeds and disturbs disturbed soil margins
- Syang village monastery gardens: ancient walnut and apple trees maintained by Nyingma monks
- Apple brandy ('Marpha ko raksi') available throughout Jomsom — produced in Marpha and sold as the signature Mustang product; Jomsom is primary distribution point
- Muktinath pilgrimage demand (200,000+/yr) creates significant fresh food market along the corridor; Jomsom serves as primary supply hub

Planetary Health — Key Themes

① Urbanization & Pollution

The completion of the Beni–Jomsom–Kora La highway (2015) transformed Jomsom from a trekking-accessible district capital into a mass-transit hub. Domestic pilgrimage traffic to Muktinath now exceeds 200,000 pilgrims annually — most transiting Jomsom by jeep in a single day. Trekkers completing the Annapurna Circuit fly out of Jomsom Airport. The town simultaneously serves backpackers, Indian pilgrims, Upper Mustang trekkers, government officials, army personnel, and local Thakali families.

This convergence has driven rapid concrete construction along the main road and riverbank — incompatible with traditional Thakali stone-and-mud architecture. Infrastructure cannot match demand: sewage treatment is absent, with waste flowing into the Kali Gandaki. The tourist market has displaced much of the town's original character. Jomsom's main street now runs the full range from Thakali guesthouses with wood-fired stoves to concrete hotels with satellite TV.

Water & Waste Pollution: No sewage infrastructure — raw waste enters the Kali Gandaki. Increasing guesthouse density along the riverbank raises fecal contamination risk for downstream communities and Shaligram fossil-collecting pilgrims. Solid waste management is entirely inadequate for peak-season visitor volumes.

Wind & Dust Hazard: Daily Kali Gandaki afternoon winds (regularly 40–60km/h+) generate intense dust storms across the riverbed — a significant respiratory hazard for residents, construction workers, and trekkers. Chronic dust exposure in an altitude-adapted community with no pollution monitoring represents an under-recognised NCD risk factor.

Visitor Type		Volume (est.)
Annapurna trekkers	Circuit	~130,000+/yr (2019)
Upper trekkers	Mustang	7,200/yr (2019)
Muktinath pilgrims (transit)		200,000+/yr
Airport passengers (Pokhara)		~30,000–40,000/yr

Visitor pressure on Jomsom infrastructure — designed for a town of ~1,370



Jomsom Airport (IATA: JMO, 2,736m) — morning-only STOL operations before valley winds close

② Environment, Ecosystem & Biodiversity

Jomsom occupies a pivotal ecological position: the southern threshold of the Trans-Himalayan cold desert, and the northern limit of Mustang's mixed broad-leaved forests. The Kali Gandaki gorge corridor is globally significant as a raptor migration highway. The Shaligram fossil beds — a

200-million-year-old Tethys Sea geological record — are the world's sole source of these ammonite stones sacred to Hindus and Buddhists.

Land degradation pressures are increasing: fuelwood collection from juniper and pine woodland; overgrazing by livestock (intensified by reduced transhumance management since trade collapse); road construction disturbing fragile Spiti shale slopes; and gravel mining from the Kali Gandaki riverbed for construction. The latter is particularly damaging — it directly reduces fossil-bearing shale exposure and alters river morphology, threatening both heritage and flood risk downstream.

Dhumba Lake: A sacred turquoise glacial lake 5km north of Jomsom at the base of Nilgiri. Fed by Nilgiri glacial meltwater and protected by community prayer flags. As Nilgiri glaciers retreat under accelerating warming, Dhumba Lake faces long-term water insecurity — an ecological and spiritual loss with no easy mitigation.

Syang & Thini Forest: Forest patches near these ancient villages provide the only woodland wildlife habitat in the immediate Jomsom area. Musk deer, blue sheep, and forest birds have been documented. These forests are under pressure from fuelwood demand and tourist foot traffic.

③ Climate Change, Hazards & Resilience

➔ Aviation Hazard (SYSTEMIC)

- Jomsom Airport (IATA: JMO, 2,736m) operates VFR-only; morning-only window before valley winds close operations
- 7 major aviation accidents on Pokhara–Jomsom route since 1970 — pattern of CFIT in deteriorating weather (1998: 18 killed; 2012: 15 killed; 2016: 23 killed; 2022: 22 killed)
- No instrument approach procedures; no formally documented escape route for Jomsom-bound flights unable to land
- 90% of Nepal aviation accidents since 1960: CFIT in cloud/mountains — Jomsom route exemplifies systemic national risk
- Weather closures cascade: medical evacuations delayed, Mustang supply chains disrupted, district isolated
- TAWS inhibition by pilots: documented in 2022 fatal crash; systemic safety culture and regulatory gap

Flood, Erosion & Earthquake

- 2015 earthquake: 15–20 riverside buildings damaged; progressive slope failure; airport temporarily closed
- Kali Gandaki bank erosion accelerating — riverside construction encroaching on active flood zone
- Gravel mining from riverbed weakening natural flood buffer and exposing infrastructure foundations
- Upstream glacial retreat increasing GLOF risk to Kali Gandaki headwaters; Dhumba Lake dam-break scenario
- Active tectonic zone — Mustang District in high seismic hazard category; building standards inconsistently applied
- 2023 Kagbeni flood (upstream) — signals increasing extreme precipitation events reaching the corridor

Wind, Dust & Cold

- Kali Gandaki afternoon wind (daily, 40–60km/h+): respiratory dust exposure for residents and visitors
- January minimum −14°C; traditional stone houses have superior thermal mass — modern concrete paradoxically warmer outside but colder inside; increasing heating energy demand
- Wind shapes all agriculture (crop damage), construction (materials transport impossible), outdoor commerce (market stalls close at noon)
- Extreme diurnal temperature swings (>20°C daily range) stress building materials and human thermoregulation
- High UV (2–3× sea-level intensity): cataracts and skin cancer risk for outdoor workers and pilgrims

Water Scarcity & Drought

- Nilgiri and Dhaulagiri glacier retreat reducing Kali Gandaki baseflow — Dhumba Lake recharge at risk
- Annual precipitation ~260–323mm: locally increasing trend but highly variable; single failed monsoon = water crisis
- 200,000+ Muktinath pilgrims competing for water resources in an already-scarce zone
- Hailstorms increasingly frequent — can destroy entire barley and buckwheat harvests within minutes
- Declining snowpack reduces spring irrigation supply for Thini, Syang, and Jomsom terrace farms
- No crop insurance; no agricultural extension services; farmers absorb entire climate risk individually

Indigenous Resilience Systems

- Thakali hospitality tradition — the 'Thali' as cultural institution: organic multi-course meals from locally grown produce; Thakali guesthouses as both economic engine and living cultural heritage. Hand-eating tradition, communal cooking, and seed-saving practices maintain food sovereignty
- Traditional wind-responsive architecture — low-profile stone buildings, internal courtyards, flat roofs with stone weights, and enclosed alleys (best exemplified at Marpha, 6km south) represent centuries of climate-adapted building intelligence; being replaced by thermally inferior concrete
- Kar-nak crop rotation — traditional fallow-rotation system (naked barley / buckwheat alternation) maintains soil fertility in nutrient-poor high-altitude soils without external inputs
- Syang Gompa (Nyingma) and Ani Gompa (Buddhist nunnery) — two active monasteries 30 minutes' walk from Jomsom; Dhekep festival (Oct/Nov) maintains ritual calendar and community cohesion. Jomsom Bon Monastery (est. ~1938, affiliated to Lubra Gompa) maintains pre-Buddhist tradition adjacent to town
- Yartung festival (September) — annual week-long horse-racing festival near Muktinath; Tibetan-influenced cultural celebration featuring horse-and-man races; draws the diaspora home and reinforces social bonds across Panchgaon communities; key event for community mental health and collective identity
- Mustang Eco Museum (est. 1992) — documents biodiversity, Buddhist monastery models, herbarium, traditional dress, and cultural heritage; educational resource and tourism anchor serving community memory function
- Community aviation safety awareness — local trekking agencies and guesthouses provide informal safety briefings to pilgrims about Jomsom Airport's morning-only operations and cancellation norms; unofficial community knowledge system compensating for institutional gaps

Health Profile

Jomsom's health burden is distinctive: it serves simultaneously as a small Thakali community in epidemiological transition and as the sole hospital district for all of Mustang — a 14,452-person district with no other hospital across 3,573 km². The Jomsom District Hospital is the health lifeline for both the 50km+ corridor south-to-north and for emergency cases from Upper Mustang and Muktinath. Its capacity is critically under-matched to its catchment.

The traditional Thakali population is shifting from infectious disease to NCDs driven by road-access dietary change and reduced physical activity. Simultaneously, Jomsom's position as a transit hub for 200,000+ Muktinath pilgrims — many elderly, arriving rapidly from sea level — creates a high-volume acute mountain sickness and HAPE presentation load that the single district hospital cannot adequately absorb.

Public Health Highlights: AMS affects 25–40% of rapid-ascent Muktinath pilgrims (3,760m); HAPE risk at Jomsom itself (2,720m) for lowland visitors. No sewage — fecal contamination of Kali Gandaki documented. UV 2–3× sea-level intensity. Daily wind-driven dust: respiratory risk. Aviation-related mass-casualty risk — seven fatal accidents since 1970. Hospital is district lifeline — when weather closes airport, emergency evacuations require military helicopter.



Jomsom District Hospital — only hospital serving all of Mustang District (14,452 people across 3,573 km²). Under-resourced for its catchment and the 200,000+ annual Muktinath pilgrims.

▲ Altitude & AMS/HAPE

💧 Water Quality & Zoonotic Risk

Jomsom at 2,720m: moderate AMS risk for lowland visitors arriving by jeep in hours. Muktinath at 3,760m: AMS affects 25–40% of rapid-ascent pilgrims — virtually all arriving without acclimatisation. HAPE is a medical emergency risk. Thakali and Tibetan-Bhote populations carry EPAS1 genetic altitude adaptation ('super-athlete gene'). The District Hospital is the only hospital within 50km+; helicopter from Pokhara is the emergency alternative when airport is open.	No sewage infrastructure — raw waste enters Kali Gandaki from guesthouses and the main market. No systematic water quality monitoring. Pilgrims bathing in Kali Gandaki (Shaligram collection) face fecal contamination risk. Yak and livestock community maintains human-animal interface. Open border with Tibet creates cross-border disease pathway. Mass pilgrim gatherings at Muktinath (108-spout bathing) create congregation-based infectious disease risk.
 Non-Communicable Disease Post-road dietary change driving NCD risk factors: hypertension, obesity, type 2 diabetes rising across all age groups. Processed food and refined sugar from Indian and Chinese road traders displacing traditional Thakali diet (organic grains, legumes, yak dairy). Daily wind-driven dust exposure: unmonitored chronic respiratory disease risk. UV 2–3× sea-level: cataracts and skin cancer elevated for outdoor market workers and trekking guides. Cardiovascular disease and COPD lead nationally; trend mirrored locally.	 Mental Health Depression 11–17% nationally; treatment gap >90% mirrored locally. Aviation disaster trauma — the Jomsom corridor has seen 7 fatal crashes since 1970; local guides, airport staff, and army personnel are repeatedly first responders to mass-casualty events with no formal psychological support. Rapid cultural disruption — commercialisation of the Thakali hospitality tradition, diaspora, identity loss — drives community grief. Yartung festival, Syang monasteries, and community guesthouses remain key protective factors. (Luitel et al., 2017; 2018)

Cross-Cutting Themes

GEDSI Growing proportion of female-headed households as Thakali male out-migration accelerates (>80% of Thakalis now outside Thak Khola). Women manage guesthouses, farms, and community life during extended male absence. Literacy gap: male higher than female. Post-road in-migration has introduced new social hierarchies alongside traditional Thakali clan structures. Disability and elderly face acute healthcare barriers — single district hospital with limited capacity. Government-posting in-migrants (army, police, civil servants) from varied ethnic backgrounds create multilayered social dynamics. GEDSI lens essential for equitable tourism economy planning, disaster preparedness, and health service delivery.	Mental Health & Wellbeing Jomsom carries a distinctive and largely unacknowledged trauma burden: 7 aviation disasters on its approach corridor since 1970, including 22 killed in 2022 and 23 killed in 2016. Local trekking guides, army personnel, and airport staff repeatedly respond to mass-casualty events with no formal psychological debriefing or support. This aviation disaster PTSD operates alongside the standard disruption of rapid cultural commercialisation and Thakali diaspora grief. Treatment gap >90% mirrors national trend. Yartung festival and Syang monastery community structure remain primary protective factors. Thakali hospitality culture ('warm and friendly nature') is itself a resilience resource and should be recognised as such in any wellbeing planning.	Culture, Heritage & Human–Nature Jomsom is the living centre of Thakali culture: the last significant Thakali community in situ, with the cuisine, monastery tradition, Yartung festival, and Panch Gaun village cluster (Thini, Syang, Marpha, Chhairo, Chimang) as living heritage assets. The Kali Gandaki Shaligram fossil beds encode a 200-million-year human–river–geological relationship; the daily collection of sacred stones by pilgrims represents one of the world's longest-continuing geological pilgrimage traditions. Dhumba Lake as a glacially-fed sacred site embodies the human–climate relationship: its health is inseparable from Nilgiri glacier health. The Mustang Eco Museum (1992) is a critical community memory institution and should be resourced accordingly.
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Sources

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